

Chapter 7: Chemical Bonding

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CHM111
Lecture 7



Part I

Models of Chemical Bonding

Outline of Part I: Models of Chemical Bonding

Why Do Atoms Bond?

Outline of Part I: Models of Chemical Bonding

Why Do Atoms Bond?

Why Do Atoms Form Bonds?

The Fundamental Driving Force

Atoms bond because bonding **lowers the potential energy** of the system — the attractive forces between positive and negative particles outweigh the repulsive forces.

Three Models of Chemical Bonding

- ▶ **Ionic bonding:** transfer of one or more electrons from a metal to a nonmetal
- ▶ **Covalent bonding:** sharing of electrons between two nonmetals
- ▶ **Metallic bonding:** pooling of valence electrons among many metal atoms

The type of bond formed depends on the **atomic properties** of the bonding partners — especially their electronegativity and ionization energy.

Atomic Properties: Metals vs. Nonmetals

Key Property Comparison

Property	Metal atom	Nonmetal atom
Atomic size	larger	smaller
Z_{eff} (eff. nuclear charge)	lower	higher
Ionization Energy	lower	higher
Electron Affinity	less negative	more negative
“Wants” electrons	less	more

Metals tend to occupy the **lower-left** of the periodic table; nonmetals occupy the **upper-right**. The larger the difference in these properties, the more **ionic** the bond.

Check Your Understanding

Without looking at a periodic table, predict which type of bond (ionic, covalent, or metallic) would form between each pair:

- (a) Na and Cl
- (b) H and O
- (c) Fe and Fe

Come to lecture to see the answer.

Part II

Ionic Bonding

Outline of Part II: Ionic Bonding

Ionic Bonding

Born-Haber Cycle

Outline of Part II: Ionic Bonding

Ionic Bonding

Born-Haber Cycle

What Is Ionic Bonding?

Ionic Bond Formation

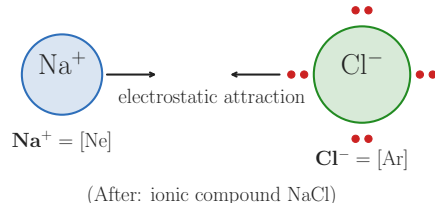
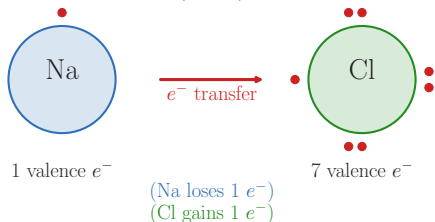
An **ionic bond** forms when one or more electrons are **transferred** from a metal atom to a nonmetal atom. The resulting oppositely charged ions attract each other.

Sodium Chloride: Na^+ and Cl^-

- ▶ Na: $[\text{Ne}] 3s^1 \rightarrow \text{Na}^+: [\text{Ne}]$ (loses $1 e^-$)
- ▶ Cl: $[\text{Ne}] 3s^2 3p^5 \rightarrow \text{Cl}^-: [\text{Ne}] 3s^2 3p^6 = [\text{Ar}]$ (gains $1 e^-$)
- ▶ Both ions achieve **noble-gas electron configurations**

Ionic Bond Formation: $\text{Na} + \text{Cl}$

(Before)



The Octet Rule

Statement of the Octet Rule

When atoms bond, they tend to **gain, lose, or share electrons** to attain a filled outer level of **8 electrons** (an “octet”).

- ▶ Exception: **H, He, Li, Be** — satisfied with 2 electrons (“duet”)
- ▶ Exception: expanded octets for elements in Period 3 and beyond (e.g., PCl_5 , SF_6)

Ionic Bonding Satisfies the Octet Rule

- ▶ $\text{Na}^+ = [\text{Ne}] \Rightarrow 8 \text{ valence } e^- \text{ in outer shell}$
- ▶ $\text{Cl}^- = [\text{Ar}] \Rightarrow 8 \text{ valence } e^- \text{ in outer shell}$
- ▶ $\text{Mg}^{2+} = [\text{Ne}]; \text{O}^{2-} = [\text{Ne}]$ — both noble-gas configs

The octet rule is a **guideline**, not an absolute law — we will see exceptions in Lewis structures.

NaCl: Crystal Structure

3D Crystalline Solid

- ▶ NaCl does **not** consist of discrete molecules
- ▶ It forms an extended **crystal lattice**: each Na^+ surrounded by 6 Cl^- , each Cl^- by 6 Na^+
- ▶ “NaCl” is an **empirical formula** — the simplest whole-number ratio of ions in the crystal
- ▶ The repeating unit is called a **formula unit**

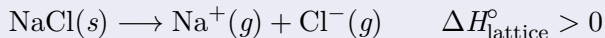
Ionic Solid Properties

- ▶ Hard and rigid (ions locked in place by strong electrostatic forces)
- ▶ Brittle (layers can slip, bringing like charges together)
- ▶ High melting & boiling points
- ▶ Conduct electricity only when melted or dissolved

Lattice Energy

Definition

Lattice energy (LE) is the energy required to separate 1 mol of an ionic solid into its gaseous ions:



Factors Affecting Lattice Energy

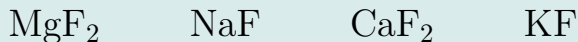
- ▶ **Higher ionic charge** $\Rightarrow$ larger LE
- ▶ **Smaller ionic radius** $\Rightarrow$ larger LE
- ▶ Coulomb's law: $F \propto \frac{q_1 q_2}{r^2}$

Examples (kJ/mol)

Compound	LE
LiF	1050
NaCl	786
MgO	3795
CaO	3414

Check Your Understanding

Rank these ionic compounds in order of **increasing** lattice energy (smallest to largest):



(Hint: consider ionic charge and size.)

Come to lecture to see the answer.

Outline of Part II: Ionic Bonding

Ionic Bonding

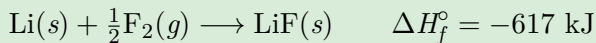
Born-Haber Cycle

The Born-Haber Cycle: Overview

What Is It?

The **Born-Haber cycle** applies Hess's Law to ionic compound formation. It breaks the overall reaction into measurable steps to calculate the lattice energy, which cannot be measured directly.

Overall Reaction for LiF Formation



Hess's Law: $\Delta H_f^\circ = \Delta H_{\text{sub}} + \Delta H_{\text{bond}} + \text{IE} + \text{EA} + \Delta H_{\text{lattice}}^\circ$

We know all terms except LE $\Rightarrow$ we can calculate it!

Born-Haber Cycle for LiF

Step	ΔH (kJ)
$\text{Li}(s) \rightarrow \text{Li}(g)$ (sublimation)	+161
$\frac{1}{2}\text{F}_2(g) \rightarrow \text{F}(g)$ (bond dissociation)	+79.5
$\text{Li}(g) \rightarrow \text{Li}^+(g) + e^-$ (IE_1)	+520
$\text{F}(g) + e^- \rightarrow \text{F}^-(g)$ (EA)	-328
$\text{Li}^+(g) + \text{F}^-(g) \rightarrow \text{LiF}(s)$ (LE)	-1050
Overall (sum)	-617

Why is LiF a stable compound?

LiF (s) forms and is stable because its enthalpy of formation is negative.

$|\Delta H_{\text{lattice}}^\circ|$ (1050 kJ) > total energy cost ($\text{IE} + \text{sub} + \text{bond}$) - EA $\approx$ 433 kJ.

433 kJ of energy is required to form the Li and F ions, but since the lattice energy (energy released when they come together) is larger in magnitude, the overall formation is favorable.

Why Ionic Solids Exist

The Energy Argument

Ionic solids form only because:

$$|\Delta H_{\text{lattice}}^{\circ}| > \sum \text{IE} + \sum \text{EA (endothermic part)}$$

- ▶ For LiF: $1050 \text{ kJ} > 520 + (-328) = 192 \text{ kJ}$ (of net ionization cost)
- ▶ The process of forming gaseous ions is endothermic overall
- ▶ The **lattice energy released** when ions pack together into a crystal is what drives the reaction

Key Insight

Without the enormous lattice energy stabilization, ionic compounds would **not form spontaneously**. This is why only certain metal-nonmetal combinations produce stable ionic solids.

Check Your Understanding

Using the Born-Haber steps for LiF:

If the lattice energy of LiF were only -400 kJ instead of -1050 kJ, would LiF(s) still be stable? Explain your reasoning.

Come to lecture to see the answer.

Part III

Covalent Bonding

Outline of Part III: Covalent Bonding

Covalent Bonding

Metallic Bonding

Outline of Part III: Covalent Bonding

Covalent Bonding

Metallic Bonding

Covalent Bonding: Sharing Valence Electrons — (Co $\implies$ Sharing)

A **covalent bond** forms when two atoms **share** one or more pairs of electrons. This typically occurs between **two nonmetals** (or a nonmetal and a metalloid).

Why Share?

- ▶ Both atoms have **high IE**
(hold onto their electrons tightly)
- ▶ Both have **high EN**
(strongly attract electrons)
- ▶ Neither atom fully takes e^- from the other
- ▶ Sharing allows both atoms to “feel” an octet

Simple Examples

- ▶ H_2 : $H-H$ (single bond, share 1 pair)
- ▶ O_2 : $O=O$ (double bond, share 2 pairs)
- ▶ N_2 : $N\equiv N$ (triple bond, share 3 pairs)
- ▶ H_2O : each $O-H$ is a single bond

⚠ Not all covalent compounds are molecular compounds! (see below)

 **Covalent molecular compounds** are described by **molecular formulas** giving the *actual* number of atoms in each molecule.

Covalent network compounds are described by **formula units** giving the ratio of atoms in the lattice.

Bond Order, Bond Energy, and Bond Length

Definitions

- ▶ **Bond order**: number of bonding pairs shared between atoms
(1 = single, 2 = double, 3 = triple)
- ▶ **Bond energy (BE)**: energy required to break 1 mol of a given bond in the gas phase (always positive — breaking bonds requires energy)
- ▶ **Bond length (BL)**: distance between bonded nuclei at equilibrium

As bond order increases:

↑ Bond Energy (harder to break)

↓ Bond Length (nuclei pulled closer)

∴ Triple bonds are the **strongest** and **shortest**.

Check Your Understanding

Without looking at a table, predict the trend:

Which bond is **longer** and which requires **more energy** to break?

(a) C–O or C=O

(b) N–N or N≡N

Come to lecture to see the answer.

Covalent vs. Network Covalent Substances

Molecular Covalent

- ▶ Discrete molecules (H_2O , CO_2 , glucose)
- ▶ **Soft**, low melting/boiling points
- ▶ Interactions between molecules are weak (intermolecular forces)
- ▶ Do **not** conduct electricity

Network Covalent

- ▶ Covalent bonds form an extended network (SiO_2 , diamond)
- ▶ **Hard**, high melting/boiling points
- ▶ Extremely strong — breaking the solid requires breaking covalent bonds
- ▶ Do **not** conduct electricity (generally)

Sand (SiO_2) and diamond (C) are network covalent — no discrete molecules exist. That's why they are **extremely hard** despite being covalent.

Outline of Part III: Covalent Bonding

Covalent Bonding

Metallic Bonding

Metallic Bonding: The Electron-Sea Model

Description

In a metal, **many atoms pool their valence electrons** into a shared “sea” of delocalized electrons. Metal cations are held together by the attraction to this mobile electron cloud.

Why Delocalized?

- ▶ Core electrons **shield** valence electrons from the nucleus
- ▶ Low IE: valence electrons are loosely held
- ▶ Electrons flow freely from atom to atom $\Rightarrow$ **electrical conductivity**

Properties Explained

- ▶ **Conducts electricity & heat:** mobile electrons carry energy and charge
- ▶ **Malleable/ductile:** cations can shift without breaking bonds
- ▶ **Lustrous:** electrons absorb and re-emit light
- ▶ **Moderately low MP, very high BP:** hard to completely separate cations from electrons

Part IV

Lewis Symbols and Structures

Outline of Part IV: Lewis Symbols and Structures

Lewis Symbols and Structures

Outline of Part IV: Lewis Symbols and Structures

Lewis Symbols and Structures

Lewis Electron-Dot Symbols

Definition

A **Lewis symbol** shows the element symbol surrounded by dots representing its **valence electrons**. One dot per electron; pair up after all four sides have one.



1 val. e^-



2 val. e^-



1 val. e^-



2 val. e^-



3 val. e^-



4 val. e^-



5 val. e^-



6 val. e^-



7 val. e^-



8 val. e^-

Each ● represents one valence electron.
Paired dots = lone pair; unpaired dots tend to form bonds.

For **main-group** elements, the number of valence electrons (dots) corresponds to group number: (Group 1: 1 dot, Group 17: 7 dots, Group 18: 8 dots)

The dot arrangement predicts **bonding behavior** — unpaired dots tend to form bonds.

Drawing Lewis Structures: The Rules

Step-by-Step Procedure

1. Count total valence electrons (add for anions, subtract for cations)
2. Write the **skeletal structure**: connect atoms with single bonds; least electronegative atom is usually central
3. Subtract electrons used in bonds ($2 \times$ bonds)
4. Complete octets on **terminal** (outer) atoms first; H gets $2 e^-$
5. Place remaining electrons on the **central** atom
6. If central atom lacks octet: convert lone pairs to **multiple bonds**

A **lone pair** (nonbonding pair) is a pair of electrons not shared between atoms. A **bonding pair** is shared.

Example: Lewis Structure of H₂O

Water

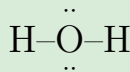
Draw the Lewis structure of water.

Step-by-Step

1. Valence e^- : $O(6) + 2 \times H(1) = 8$
2. Skeletal: H–O–H (O is central)
3. Used in 2 bonds: $2 \times 2 = 4 e^-$ used
4. Remaining: $8 - 4 = 4 e^- = 2$ lone pairs on O
5. Each H has 2 e^- (duet ✓); O has 8 e^- ✓

Result

Ö surrounded by two lone pairs;



2 bonding pairs + 2 lone pairs on O

O achieves an octet; each H achieves a duet.

Example: Lewis Structure of CO₂

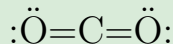
Carbon Dioxide

Draw the Lewis structure of CO₂.

Step-by-Step

1. Valence e^- : $C(4) + 2 \times O(6) = 16$
2. Skeletal: O—C—O (C central, least EN)
3. 2 bonds use 4 e^- ; remaining: 12
4. Complete O octets: 3 lone pairs each = 12 e^- ; none left for C
5. C has only 4 e^- — no octet!
Convert lone pairs to double bonds
6. O=C=O: C now has 8, each O has 8

Result



Two double bonds; each O has 2 lone pairs

CO₂ is **linear**; the two double bonds give C its full octet.

Check Your Understanding

Draw the Lewis structure for NH_3 (ammonia).

- (a) How many valence electrons total?
- (b) How many bonding pairs?
- (c) How many lone pairs on N?

Come to lecture to see the answer.

Exceptions to the Octet Rule

Electron-Deficient Molecules

Some atoms form stable compounds with **fewer than 8 electrons**:

- ▶ BF_3 : B has only 6 e^- — very reactive Lewis acid
- ▶ BeCl_2 : Be has only 4 e^-
- ▶ H: only needs 2 e^-

Expanded Octets

Elements in **Period 3+** can have more than 8 electrons:

- ▶ PCl_5 : P has 10 e^- (5 bonds)
- ▶ SF_6 : S has 12 e^- (6 bonds)
- ▶ XeF_4 : Xe has 12 e^- (4 bonds + 2 lone pairs)

Expanded octets are **only possible** for elements with available d orbitals (Period 3 and beyond). Never for C, N, O, F.

Part V

Formal Charges and Resonance

Outline of Part V: Formal Charges and Resonance

Formal Charges and Resonance

Outline of Part V: Formal Charges and Resonance

Formal Charges and Resonance

Formal Charges

Definition and Formula

The **formal charge** on an atom in a Lewis structure is the hypothetical charge the atom would have if all bonding electrons were shared *equally*:

$$FC = (\text{valence } e^-) - (\text{lone pair } e^-) - \frac{1}{2}(\text{bonding } e^-)$$

Rules for Best Lewis Structure

- ▶ Prefer structures where all formal charges are **zero**
- ▶ If charges unavoidable: **negative** FC on the **more electronegative** atom
- ▶ Minimize the **magnitude** of formal charges
- ▶ The sum of all FC must equal the overall charge of the molecule/ion

Example: Formal Charges in CO₂ vs. CO

CO

Calculate formal charges in carbon monoxide C≡O (triple bond, lone pair on each).

C in C≡O

$$FC_C = 4 - 2 - \frac{1}{2}(6) = 4 - 2 - 3$$

$$FC_C = -1$$

O in C≡O

$$FC_O = 6 - 2 - \frac{1}{2}(6) = 6 - 2 - 3$$

$$FC_O = +1$$

Sum of FC: $(-1) + (+1) = 0$ ✓ (CO is neutral). The structure with the triple bond is preferred because it gives lower formal charges than a double bond would.

Check Your Understanding

For the molecule NO_2^- (nitrite ion), two Lewis structures are possible. Which is better?

Structure A: N has double bond to one O, single bond to other (all octets satisfied with one lone pair on N).

Hint: calculate FC on N for both and pick the one with FC closer to zero.

Come to lecture to see the answer.

Resonance Structures

When One Lewis Structure Is Not Enough

Sometimes, two or more valid Lewis structures can be drawn for the same molecule. These are called **resonance structures**.

- ▶ Neither structure alone is correct — the **actual molecule** is a **hybrid** of all resonance structures
- ▶ Double-headed arrow ($\leftrightarrow$) between resonance structures
- ▶ Electron positions differ, but **atom positions** do not change

Ozone O_3

Two resonance structures: $\text{O}=\text{O}-\ddot{\text{O}}: \leftrightarrow :\ddot{\text{O}}-\text{O}=\text{O}$

The actual O_3 molecule has **both bonds equal** in length and energy — between single and double bond character. **Bond order = 1.5**.

Resonance Hybrid: Benzene and Beyond

Nitrate Ion NO_3^-

Three equivalent resonance structures, differing in which N–O bond is the double bond.

Actual molecule: all three N–O bonds are equal (bond order = 1.33).

Benzene C_6H_6

Two resonance structures (alternating single/double bonds).

Actual: all six C–C bonds are identical (≈ 1.5 bond order). Benzene is unusually stable due to delocalization.

Common Misconception

The molecule does **not** “switch” between resonance structures. The actual electron distribution is a **single state** that is spread (delocalized) over all resonance structures simultaneously.

Part VI

Bond Energies and Polarity

Outline of Part VI: Bond Energies and Polarity

Strengths of Ionic and Covalent Bonds

Electronegativity and Bond Polarity

Outline of Part VI: Bond Energies and Polarity

Strengths of Ionic and Covalent Bonds

Electronegativity and Bond Polarity

Bond Energy and Enthalpy of Reaction

Using Bond Energies to Estimate $\Delta H^\circ_{\text{rxn}}$

$$\Delta H^\circ_{\text{rxn}} \approx \sum BE_{\text{bonds broken}} - \sum BE_{\text{bonds formed}}$$

- ▶ Breaking bonds: **endothermic** (requires energy, positive contribution)
- ▶ Forming bonds: **exothermic** (releases energy, negative contribution)

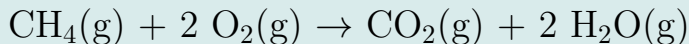
Example: $\text{H}_2 + \text{Cl}_2 \rightarrow 2 \text{HCl}$

$$\begin{aligned}\Delta H^\circ &= [BE(\text{H-H}) + BE(\text{Cl-Cl})] - [2 \times BE(\text{H-Cl})] \\ &= [436 + 243] - [2 \times 432]\end{aligned}$$

$$\Delta H^\circ \approx -185 \text{ kJ}$$

Check Your Understanding

Use average bond energies to estimate ΔH° for:



Bond energies (kJ/mol): C–H = 414, O=O = 498, C=O = 799,
O–H = 464

Come to lecture to see the answer.

Outline of Part VI: Bond Energies and Polarity

Strengths of Ionic and Covalent Bonds

Electronegativity and Bond Polarity

Electronegativity

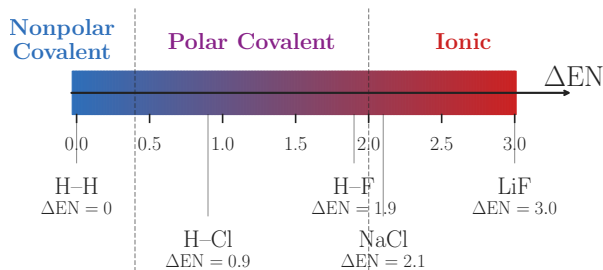
Definition

Electronegativity (EN) is the relative ability of a bonded atom to attract the shared electrons toward itself. (Pauling scale: F = 4.0, most electronegative.)

Periodic Trends in EN

- ▶ **Increases** across a period (left → right): higher Z_{eff} pulls shared electrons closer
- ▶ **Decreases** down a group: electrons farther from nucleus, more shielded
- ▶ Highest: **F** (4.0) — upper right
- ▶ Lowest: **Cs** — lower left
- ▶ $\text{EN} \propto \frac{1}{\text{atomic radius}}$

Bond Polarity and Electronegativity Difference



The **type of bond** is determined by ΔEN , the difference in bound particles' electronegativity values. (+ and - ions clearly have the greatest ΔEN .)

Bond Polarity

Nonpolar, Polar, and Ionic Bonds

- ▶ $\Delta\text{EN} < 0.4$: **Nonpolar covalent** — electrons shared equally (H–H, C–C)
- ▶ $0.4 \leq \Delta\text{EN} < 2.0$: **Polar covalent** — unequal sharing; partial charges δ^+/δ^-
- ▶ $\Delta\text{EN} \geq 2.0$: **Ionic** — electron transfer (LiF: $\Delta\text{EN} = 3.0$)

Examples

Bond	EN ₁	EN ₂	ΔEN
H–H	2.1	2.1	0.0 (nonpolar covalent)
H–Cl	2.1	3.0	0.9 (polar covalent)
H–F	2.1	4.0	1.9 (polar covalent)
Na–Cl	0.9	3.0	2.1 (ionic)
Li–F	1.0	4.0	3.0 (ionic)

Dipole Moments

Quantifying Bond Polarity

A **dipole moment** (μ) arises in a polar bond: the partial positive charge (δ^+) and partial negative charge (δ^-) separated by a distance d .

$$\mu = Q \cdot d$$

Units: **debye (D)**; $1 \text{ D} = 3.336 \times 10^{-30} \text{ C}\cdot\text{m}$

Molecular Dipole Moment

The **molecular dipole moment** is the **vector sum** of all bond dipoles. A molecule can have polar bonds yet be **nonpolar overall** if the bond dipoles cancel.

A molecule with:

► a large partial-charge difference: (δ^+) and (δ^-)

► and/or a long bond length

will generally have a larger dipole moment (more Debye) and interact more strongly with electric fields and other polar molecules.

CO₂ (linear): two C=O dipoles point in **opposite directions** $\Rightarrow$ cancel $\Rightarrow$ **nonpolar molecule**

H₂O (bent): two O–H dipoles do **not** cancel $\Rightarrow$ **polar molecule** ($\mu = 1.85 \text{ D}$)

Percent Ionic Character

What Does It Mean?

Most bonds are **neither purely covalent nor purely ionic**. Instead, electrons are shared *unequally*.

Percent ionic character tells us how much charge separation a bond has compared with a **completely ionic bond**.

$$\% \text{ ionic character} = \frac{\mu_{\text{measured}}}{\mu_{\text{ionic}}} \times 100$$

where

$$\mu_{\text{ionic}} = e \cdot d$$

is the dipole moment if one atom carried a full $+1e$ charge and the other a full $-1e$ charge.

$e = 1.60 \times 10^{-19}\text{C}$ and represents the charge of an electron.

Interpreting the Value

- ▶ **0%** → perfectly nonpolar covalent
- ▶ **100%** → completely ionic
- ▶ Most polar covalent bonds fall somewhere between these extremes.

E.g. if a bond has 35% ionic character, its measured dipole moment is only

$$0.35 \times (e \cdot d),$$

meaning the atoms behave approximately as if they carried charges of

$$\pm 0.35e.$$

Exam-Style Example: Dipole Moments

Example 1: Calculate a Dipole Moment

A diatomic molecule has partial charges of $\pm 0.35e$ separated by 180 pm. Calculate the dipole moment.

Solution

$$Q = (0.35)(1.602 \times 10^{-19} \text{ C})$$

$$d = 180 \times 10^{-12} \text{ m}$$

$$\mu = Qd$$

$$\mu = \frac{(0.35)(1.602 \times 10^{-19})(180 \times 10^{-12})}{3.336 \times 10^{-30}} = 3.02 \text{ D}$$

Example 2: Percent Ionic Character

A diatomic molecule has $\mu_{\text{meas}} = 3.02 \text{ D}$ and a bond length of 180 pm.

For a **100% ionic bond**,

$$\mu_{\text{ionic}} = \frac{(1.602 \times 10^{-19})(180 \times 10^{-12})}{3.336 \times 10^{-30}} = 8.64 \text{ D}$$

$$\% \text{ ionic} = \frac{\mu_{\text{meas}}}{\mu_{\text{ionic}}} \times 100 = \frac{3.02}{8.64} \times 100 = 35.0\%$$

Check Your Understanding

Identify each bond as nonpolar covalent, polar covalent, or ionic, and indicate the direction of the dipole (if any):

(a) C–H ($\Delta\text{EN} \approx 0.4$)

(b) N–O ($\Delta\text{EN} \approx 0.5$)

(c) K–F ($\Delta\text{EN} \approx 3.2$)

Come to lecture to see the answer.

Part VII

Molecular Geometry

Outline of Part VII: Molecular Geometry

Molecular Structure and Polarity

Outline of Part VII: Molecular Geometry

Molecular Structure and Polarity

VSEPR Theory: Introduction

Valence Shell Electron Pair Repulsion (VSEPR)

Electron pairs around the central atom **repel each other** and arrange themselves to **minimize repulsion**. This determines molecular geometry.

Key Principle

- ▶ Count **electron domains** (bonding pairs + lone pairs) around the central atom
- ▶ **Lone pairs** repel more strongly than bonding pairs ($LP > BP$)
- ▶ Multiple bonds count as **one** electron domain
- ▶ Electron-domain geometry determines **molecular geometry** (atom positions only)

Molecular geometry is determined by **atom positions**, not by lone pairs. But lone pairs **affect bond angles**.

Selected VSEPR Geometries

Geometry	Bonds	Angle
Linear	2	180°
Trig. planar	3	120°
Tetrahedral	4	109.5°
Trig. Pyr.	3+1 LP	$<107^\circ$
Bent	2+2 LP	$<104.5^\circ$



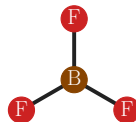
Lone pairs **compress** bond angles below ideal values.

Linear



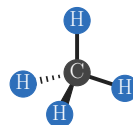
e.g. BeCl_2
 180°

Trigonal Planar



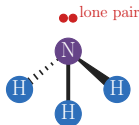
e.g. BF_3
 120°

Tetrahedral



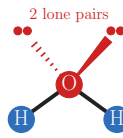
e.g. CH_4
 109.5°

Trigonal Pyramidal



e.g. NH_3
 107°

Bent



e.g. H_2O
 104.5°

Key

- Bonding pair / atom
- Lone pair
- ⋯||| Bond behind plane
- Bond in plane
- ◀ Bond in front of plane

VSEPR: Applying the Method

How to Determine Molecular Geometry

1. Draw the Lewis structure
2. Count electron domains on central atom (bonding + lone pairs)
3. Identify the **electron-domain geometry**
4. Identify the **molecular geometry** (ignore lone pairs)
5. Predict approximate bond angles

Molecule	Bonds	Lone Pairs	Mol. Geometry	Angle
BeCl ₂	2	0	Linear	180°
BF ₃	3	0	Trigonal planar	120°
CH ₄	4	0	Tetrahedral	109.5°
NH ₃	3	1	Trigonal pyramidal	107°
H ₂ O	2	2	Bent	104.5°

Check Your Understanding

Determine the molecular geometry and approximate bond angle for each:

- (a) SO_2 (S is central, has 1 lone pair, 2 bonds including one double bond)
- (b) PCl_3 (P is central, 3 bonds, 1 lone pair)
- (c) CCl_4 (C is central, 4 bonds, 0 lone pairs)

Come to lecture to see the answer.

Molecular Polarity — Determined by Net Dipole Moment

Combining Geometry and EN Differences

Whether a **molecule** is **polar** (i.e. has a net dipole moment) depends on **both**:

1. Whether the **bonds** are polar (determined by ΔEN)
2. Whether the bond dipoles **cancel** due to symmetry

Nonpolar Molecules

- ▶ CO_2 (linear): dipoles cancel
- ▶ BF_3 (trigonal planar): dipoles cancel
- ▶ CCl_4 (tetrahedral): dipoles cancel
- ▶ Any molecule where bond dipoles point symmetrically

Polar Molecules

- ▶ H_2O (bent): dipoles add up
- ▶ NH_3 (trigonal pyramidal): net dipole
- ▶ HCl (linear, only 1 bond): inherently polar
- ▶ Lone pairs break molecular symmetry

Check Your Understanding

Classify each molecule as **polar** or **nonpolar overall**:

- (a) CHCl_3 (tetrahedral; 3 C–Cl bonds and 1 C–H bond)
- (b) CS_2 (linear; two C=S bonds, symmetric)
- (c) H_2S (bent; like H_2O but S is less electronegative)

Come to lecture to see the answer.

Extended VSEPR: Five and Six Electron Domains

Trigonal Bipyramidal (5 Domains)

- ▶ **0 LP:** trigonal bipyramidal (PCl_5); axial bonds 180° , equatorial 120°
- ▶ **1 LP:** seesaw (SF_4)
- ▶ **2 LP:** T-shaped (ClF_3)
- ▶ **3 LP:** linear (XeF_2)

Octahedral (6 Domains)

- ▶ **0 LP:** octahedral (SF_6); all bonds 90°
- ▶ **1 LP:** square pyramidal (IF_5)
- ▶ **2 LP:** square planar (XeF_4)

These expanded geometries only occur for **Period 3+** central atoms that can accommodate more than 8 electrons.

The following slide is taken from the
“Molecular Shapes” handout.



You must memorize all these molecular geometries!

e ⁻ Group Arrangement (no. of groups)	 Linear (2)	 Trigonal Planar (3)		 Tetrahedral (4)			 Trigonal Bipyramidal (5)				 Octahedral (6)		
(Class) Molecular shape	 (AX ₂) Linear	 (AX ₃) Trigonal Planar	 (AX ₂ E) Bent	 (AX ₄) Tetrahedral	 (AX ₃ E) Trigonal Pyramidal	 (AX ₂ E ₂) Bent	 (AX ₅) Trigonal Bipyramidal	 (AX ₄ E) Seesaw	 (AX ₃ E ₂) T-Shaped	 (AX ₂ E ₃) Linear	 (AX ₆) Octahedral	 (AX ₅ E) Square Pyramidal	 (AX ₄ E ₂) Square Planar
No. of bonding groups	2	3	2	4	3	2	5	4	3	2	6	5	4
No. of lone pairs	0	0	1	0	1	2	0	1	2	3	0	1	2
Bond angle	180°	120°	<120°	109.5°	<109.5°	<109.5°	90° (ax.) 120° (eq.)	<90° (ax.) <120° (eq.)	<90° (ax.)	180°	90°	<90°	90°

Part VIII

Summary

Outline of Part VIII: Summary

Chapter Summary

Outline of Part VIII: Summary

Chapter Summary

Key Concepts — Chapter 7

- ▶ **Ionic bonds:** electron transfer from metal to nonmetal; driven by lattice energy (Born-Haber cycle); ionic solids are hard, brittle, high-MP.
- ▶ **Covalent bonds:** electron sharing between nonmetals; described by Lewis structures and the octet rule.
- ▶ **Lewis structures:** valence e^- count $\rightarrow$ skeletal structure $\rightarrow$ complete octets $\rightarrow$ multiple bonds as needed.
- ▶ **Formal charges:** minimize FC to pick the best resonance structure; negative FC on more electronegative atom.
- ▶ **Resonance:** delocalized electrons over multiple equivalent structures; actual molecule is the hybrid.
- ▶ **Bond energy & length:** increase in bond order $\Rightarrow$ shorter, stronger bonds.
- ▶ **Electronegativity:** drives bond polarity; ΔEN determines ionic vs. polar covalent vs. nonpolar covalent.
- ▶ **VSEPR:** electron domains repel; molecular geometry determines bond angles and whether the molecule is polar.

Key Equations and Formulas — Chapter 7

Formula	Meaning
$FC = V - LP - \frac{1}{2}B$	Formal Charge Calculation (V =valence, LP =lone pair e^- , B =bonding e^-)
$\Delta H^\circ \approx \sum BE_{\text{broken}} - \sum BE_{\text{formed}}$	Enthalpy from bond energies
$\mu = Q \cdot d$	Dipole moment (charge $\times$ distance)
$F \propto q_1 q_2 / r^2$	Lattice energy $\propto$ ionic charge, $\propto 1/r$